Two-page summary

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Toward an optimal data-generating process for simulating predictive-biomarker interactions in aggregated N-of-1 trials (pmsimstats team).

Goal

A predictive biomarker is validated by an interaction test. In an aggregated N-of-1 design every participant is observed on and off drug, so the test can be made within participants. Its power is estimated by simulation, and a simulation can only show what its data-generating process (DGP) allows. Paper 01 asks what that DGP should be.

Approach

The DGP is a multivariate normal model in which the baseline biomarker and three latent response components (drug response, placebo response, natural history) at every visit form one vector. The design assembles its correlation matrix. The predictive biomarker enters as a correlation c_{bm} with the drug response at on-drug visits (Hendrickson et al. 2020). We take the elements of the DGP one at a time, compare candidate choices for each, and judge them against three requirements:

1. **a valid joint distribution:** the correlation matrix must be positive definite;
2. **an interaction that survives carryover:** the biomarker-by-treatment interaction must remain a contrast between on-drug and off-drug visits;
3. **flexibility for stress conditions:** dropout, measurement error and a different decay form are available as options, off by default.

Pharmacological plausibility breaks ties, and every parameter is labeled as a default value, a choice made here, or uncalibrated. We derive the closed-form power of a paired-difference interaction statistic (E9) and a Schur-complement bound on the largest feasible interaction, the feasibility bound. Study 1 checks both by Monte Carlo simulation; study 2 stress-tests ten analyses under the optional features. The biomarker is assumed predictive only: it is correlated with the drug response and with nothing else (the prognostic case is paper 15). The Hybrid design is the primary case, and all simulations use the default parameter values of Hendrickson et al.

Results

Infeasible matrices. Above the feasibility bound the correlation matrix is not positive definite. A DGP can repair such a matrix or refuse it. A repaired matrix simulates a different DGP from the one stated, so the recommended DGP refuses it (requirement 1).

The biomarker-response correlation under carryover. This is the element that matters most. Under the constant rule the correlation stays at its on-drug value after first exposure. The interaction is then absent among the post-baseline visits, and power falls nearly to the test size at every positive half-life. Under the decaying rule the correlation decays at the carryover half-life. Power then stays near its no-carryover value at pharmacological half-lives of 0.1 to 0.2 week and declines smoothly at longer ones. Power depends on the rule only through the on/off correlation difference (requirement 2).

Carryover on the mean. Decaying the drug-response mean from the last on-drug value (anchored) rather than from the previous visit (compounding) makes the half-life mean what it says. Its effect on power is at most 0.006.

Serial correlation and cross-component structure. The correlation structure bounds the largest interaction the DGP can hold: 0.256 under compound symmetry and 0.474 under separable AR(1) at a serial correlation of $\rho = 0.7$. The separable form is positive definite for any visit schedule and gives an exact bound. The value 0.7 is a convention. A PTSD symptom series suggests 0.9, where the bound falls to 0.34 and the Hybrid design needs 154 participants at $c_{bm} = 0.25$ instead of 278.

Optional features. In study 2, measurement error lowered power by 0.05 to 0.09, dropout of 20% by 0.01 to 0.06, and a Weibull decay moved it by up to 0.10. The planning sample sizes assume none of these features.

Planning. At the pharmacological half-life the Hybrid design needs about 280 participants at $c_{bm} = 0.25$ and 84 at 0.45 for 80% power of E9; at a 1-week half-life, 480 and 146. The recommended analysis G10 (a random-intercept CAR(1) mixed model with an analysis half-life chosen by AIC on the model without the interaction, tested with cluster-robust CR2 standard errors) rejected the null at 0.046 to 0.053. At the E9 sample size G10 power was 0.844 to 0.936 in the 20 uncapped cells, so the closed-form rule is conservative for G10. In the Hybrid design the G10 sample size for 80% power is 0.71 to 0.83 times the E9 value.

Differences from the DGP of Hendrickson et al. A separate section lists every difference and reruns their code. As published, power fell from 0.668 without carryover to 0.126 at a 0.1-week half-life; a 4,000-replicate rerun gave 0.090. With only the correlation lines replaced by the decaying rule, power was 0.638. In their grid, 20 of 54 matrices at $c_{bm} \in \{0.3, 0.6\}$ were repaired without a warning.

The recommended DGP

A biomarker-response correlation that decays after discontinuation, carryover on the mean anchored at the last on-drug value, a separable AR(1) structure at $\rho = 0.7$ with a cross-component correlation of 0.2, a check that refuses infeasible effect sizes, and options for measurement error, the decay form and dropout. Every parameter is labeled as a default value, a choice or uncalibrated.

Conclusion

First, the biomarker-response correlation must decay after discontinuation; a constant correlation removes the interaction under any carryover. Second, the serial correlation structure sets the largest interaction the DGP can hold, and the required sample size depends strongly on ρ. Third, a matrix outside the feasibility bound should be refused rather than repaired. We propose the recommended DGP as a reference for simulation studies of these designs when the biomarker is predictive only. Calibrating the serial correlation, the drug-response SD and the symptom half-life is the next step toward an optimal DGP.